

HUBNET

Mobile Data API

[API DOCUMENTATION](#)

Version
v1.0

Effective Date
April 2025

Base URL
console.hubnet.app

Supported Networks

MTN

Telecel

AirtelTigo

big-time

Rate Limited: 5 requests per minute per endpoint

Integrate fast. Deliver data. Scale confidently.

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1 Authentication

How to authorise every API request

All Hubnet API endpoints require a **Bearer Token** passed in the HTTP **token** header. You will receive your API key from the Hubnet console at console.hubnet.app.

Required Headers

```
HTTP Headers

{
  "token" : "Bearer YOUR_API_KEY",
  "Content-Type" : "application/json"
}
```

■ **Security Note:** Never expose your API key in client-side code or public repositories. Always store it in environment variables or a server-side secrets manager.

2 Check Balance

Retrieve your current wallet balance

GET

```
https://console.hubnet.app/live/api/context/business/transaction/check_balance
```

Use this endpoint to query your Hubnet wallet balance before initiating transactions. It is good practice to verify sufficient funds before bulk data bundle operations.

Sample Response

```
check_balance · Response

{
  "status" : true,
  "balance" : 500.00,
  "currency" : "GHS"
}
```

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Make a Transaction

Deliver a data bundle to a subscriber

POST

https://console.hubnet.app/live/api/context/business/transaction/{network}-new-transaction

Replace {network} with one of: mtn, at, or big-time.

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Request Parameters

Complete payload field reference

```
{
  "phone"      : "0241234567",
  "volume"     : "2000",
  "reference"  : "TCX-ABEDC-BBADE-ABBCDD",
  "referrer"   : "0270123456",
  "webhook"    : "https://your-site.com/webhook/hubnet"
}
```

Parameter	Type	Required	Length	Description
phone	String	Yes	Exactly 10 digits	Recipient's phone number in national format, e.g. 0241234567
volume	String	Yes	1 – 100 000 (MB)	Bundle size in megabytes. Convert GB to MB by multiplying by 1000
reference	String	Yes	6 – 25 characters	Unique transaction reference; must not be reused
referrer	String	Optional	Exactly 10 digits	Buyer's phone number to receive SMS confirmation
webhook	String	Optional	Valid HTTPS URL	Your endpoint to receive real-time POST callbacks on status changes

Webhook Sample Payload

Hubnet will POST this JSON to your webhook URL on every status change.

transfer.delivered

```
{
  "event"      : "transfer.delivered",
  "status"     : true,
  "message"    : "Transfer successful",
  "code"       : 200,
  "data": {
    "status"    : "Processing",
    "date_time" : "April 17, 2025 12:26:40 AM",
    "msisdn"    : "0271277777",
    "network"   : "mtn",
    "volume"    : "1000",
    "reference"  : "webhook_test"
  }
}
```

transfer.processing

```
{
  "event"      : "transfer.processing",
  "status"     : true,
  "message"    : "Order fulfillment is in progress",
  "code"       : 200,
  "data": {
    "status"    : "processing",
    "timestamp" : "June 7, 2026 4:42:33 PM",
    "msisdn"    : "0591923218",
    "network"   : "mtn",
    "volume"    : "2000",
    "reference"  : "WAL-1780850315550-52999"
  }
}
```

■ **Volume in Megabytes:** The API expects the **volume** field in MB. To send 2 GB, pass **"2000"**. To send 500 MB, pass **"500"**.

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Response Reference

Understand every field in the API response

Success Response

Success • 200 OK

```
{
  "status"      : true,
  "reason"      : "Successful",
  "code"        : "transaction initiated successfully.",
  "message"     : "0000",
  "transaction_id" : "TXN-HUBNET-00123",
  "payment_id"   : "PSH-HUBNET-456",
  "ip_address"   : "102.89.45.12",
  "reference"    : "TCX-ABEDC-BBADE-ABBCDD",
  "data": {
    "status" : true,
    "code"   : "0000",
    "message" : "Order successfully placed."
  }
}
```

Response Field Descriptions

Parameter	Type	Required	Length	Description
status	Boolean	—	—	true = transaction accepted; false = rejected
message	String	—	4 chars	Machine-readable status code (see table below)
transaction_id	String	—	Variable	Hubnet's unique ID for this transaction — store for status checks
payment_id	String	—	Variable	Hubnet payment processor reference
reference	String	—	6–25 chars	Echoes the reference you submitted
ip_address	String	—	—	Server IP that processed the request
data.status	Boolean	—	—	Delivery-layer status confirmation
data.message	String	—	—	Human-readable outcome from the delivery layer

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Transaction Status Check

Verify whether a bundle was delivered

POST

<https://console.hubnet.app/live/api/context/business/transaction/check-transaction-status>

Query the live status of a previously submitted transaction using its reference value. Use this to confirm delivery or to reconcile orders where the original response timed out.

Status Check · POST Body

```
{
  "reference" : "TCX-ABEDC-BBADE-ABBCDD"
}
```

Status Values in Response

- **success / delivered / completed** — Bundle was delivered to the subscriber
- **pending / processing** — Transaction is still being processed
- **failed / cancelled** — Transaction did not complete

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Check Transaction Status

Universal status lookup — GET, no network prefix required

GET

<https://console.hubnet.app/live/api/context/business/transaction/check-transaction-status>

A universal, network-agnostic endpoint that mirrors the same GET pattern as `check_balance`. Pass the transaction reference as a query-string parameter — no network prefix in the URL, no request body required.

Parameter	Type	Required	Description
reference	String	Yes	The transaction reference from the original bundle request

```
GET /live/api/context/business/transaction/check-transaction-status
    ?reference=ESTORE-565688-13408

Headers:
  token: Bearer YOUR_API_KEY
  Content-Type: application/json
```

Sample Responses

Delivered • 200 OK

```
{
  "status" : true,
  "message" : "0000",
  "data": {
    "status" : "delivered",
    "message" : "Bundle delivered successfully."
  }
}
```

Pending • 200 OK

```
{
  "status" : true,
  "message" : "0000",
  "data": {
    "status" : "pending",
    "message" : "Transaction is being processed."
  }
}
```

Delivery Status Values

- **delivered / success / completed** — Bundle was delivered to the subscriber
- **pending / processing** — Transaction is still being processed
- **failed / cancelled** — Transaction did not complete

Note

The reference must be from a real submitted transaction. Auto-generated test references will return 'No transaction found'.

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Network & Volume Guide

Choosing the right network and bundle size

Supported Networks

Network Name	URL Token	Notes
MTN Ghana	mtn	Largest mobile network in Ghana
AirtelTigo Ghana	at	AirtelTigo standard bundles
big-time (AirtelTigo)	big-time	Hubnet's big-time package — AirtelTigo network
Telecel Ghana	telecel	Telecel Ghana (formerly Vodafone)

Volume Conversion Reference

Bundle Size	volume value	Example Use Case
500 MB	"500"	Light browsing / social media
1 GB	"1000"	Daily casual use
2 GB	"2000"	Standard bundle
5 GB	"5000"	Heavy streaming
10 GB	"10000"	Power user / hotspot
50 GB	"50000"	Business / tethering

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Code Examples

PHP integration — production-ready

Send a Data Bundle (PHP / cURL)

```
sendBundleHubNet() · PHP

function sendBundleHubNet(string $network, float $sizeGb, string $phone): array {
    $apiKey = 'YOUR_HUBNET_API_KEY';
    $netMap = ['MTN'=>'mtn', 'AirtelTigo'=>'at', 'Telecel'=>'telecel'];
    $hubNet = $netMap[$network] ?? null;
    if (!$hubNet) return ['success'=>false, 'message'=>'Invalid network'];

    $volumeMB = (int)($sizeGb * 1000); // Convert GB → MB
    $reference = 'TCX-' . strtoupper(substr(md5(uniqid()), 0, 12));
    $endpoint = "https://console.hubnet.app/live/api/context/business/"
        . "transaction/{$hubNet}-new-transaction";

    $payload = json_encode([
        'phone' => $phone,
        'volume' => (string)$volumeMB,
        'reference' => $reference,
    ]);

    $ch = curl_init($endpoint);
    curl_setopt_array($ch, [
        CURLOPT_RETURNTRANSFER => true,
        CURLOPT_POST => true,
        CURLOPT_POSTFIELDS => $payload,
        CURLOPT_HTTPHEADER => [
            'token: Bearer ' . $apiKey,
            'Content-Type: application/json',
        ],
        CURLOPT_TIMEOUT => 30,
    ]);
    $response = curl_exec($ch);
    $httpCode = curl_getinfo($ch, CURLINFO_HTTP_CODE);
    curl_close($ch);

    $result = json_decode($response, true);
    $success = !empty($result['status']) && $result['message'] === '0000';
    return array_merge($result ?? [], ['success' => $success]);
}
```

Check Transaction Status (PHP)

```
checkHubNetStatus() · PHP

function checkHubNetStatus(string $reference, string $network): ?bool {
    $apiKey = 'YOUR_HUBNET_API_KEY';
    $netMap = ['MTN'=>'mtn', 'AirtelTigo'=>'at'];
    $net     = $netMap[$network] ?? 'mtn';
    $url     = "https://console.hubnet.app/live/api/context/business/"
        . "transaction/{$net}-transaction-status";

    $ch = curl_init($url);
    curl_setopt_array($ch, [
        CURLOPT_RETURNTRANSFER => true,
        CURLOPT_POST           => true,
        CURLOPT_POSTFIELDS     => json_encode(['reference' => $reference]),
        CURLOPT_HTTPHEADER     => ['token: Bearer ' . $apiKey,
                                   'Content-Type: application/json'],
        CURLOPT_TIMEOUT        => 10,
    ]);
    $data = json_decode(curl_exec($ch), true);
    curl_close($ch);

    if (isset($data['message']) && $data['message'] === '0000') return true;
    $s = strtolower($data['data']['status'] ?? '');
    if (in_array($s, ['success', 'delivered', 'completed'])) return true;
    if (in_array($s, ['failed', 'cancelled'])) return false;
    return null; // Unknown / still processing
}
```

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Error Handling

Retry logic and failure recovery

Robust error handling is critical for production deployments. Follow these strategies to ensure no bundle goes undelivered:

Scenario	Recommended Action
Network / cURL Timeout	Retry up to 3 times with exponential back-off (1s, 2s, 4s). Always check status via the status-check endpoint before retrying to avoid double-delivery.
HTTP 5xx from Hubnet	Server-side issue. Wait 30 seconds and retry once. If it persists, mark the order as 'awaiting_topup' and alert support.
Code 1004 — Duplicate Reference	Your reference was already used. Generate a new unique reference and re-submit. Never reuse references across transactions.
Code 1005 — Insufficient Balance	Top up your Hubnet wallet via console.hubnet.app before retrying. Implement balance threshold alerts using the Check Balance endpoint.

Code 1007 — Rate Limited	Your integration exceeds 5 requests/minute. Implement a queue with a 12-second minimum interval between requests.
status:true but delivery unknown	Use the Transaction Status Check endpoint with the stored transaction_id or reference to confirm actual delivery.

10 Rate Limits & Best Practices

Build reliable, scalable integrations

■ **Rate Limit:** All endpoints are limited to **5 requests per minute**. Exceeding this returns status code **1007**.

Unique References

Generate a cryptographically unique reference for every transaction (e.g. TCX- + 12 random hex chars). Store it in your database before calling the API to enable idempotent retries.

Check Balance First

Query your wallet balance before high-volume batch jobs to avoid failed transactions mid-run. Set up automated alerts at your chosen threshold.

Idempotent Retries

Before retrying a timed-out request, always call the status-check endpoint to confirm whether the first attempt succeeded. This prevents double-delivery.

Webhooks for Status Updates

Supply a webhook URL in the payload to receive real-time delivery confirmations. This removes the need for polling and reduces API calls.

Log Everything

Store the full API response (transaction_id, reference, status, message) against every order in your database for debugging and customer support.

Secure Your API Key

Keep your Bearer token in server-side environment variables only. Rotate it immediately if you suspect exposure.

Need Help?

Visit the Hubnet developer portal for the latest documentation, API keys, wallet top-up, and live transaction logs.

■ **Portal:** console.hubnet.app